

Confidence Interval

A 95 Percent Confidence Interval represents:

Common Shorthand

We are
95 percent confident that
the true population
parameter lies within this
interval.

Technical Interpretation

If we repeatedly collected
samples from the same
population and calculated a 95%
confidence interval from each
sample, approximately 95% of
those intervals would contain the
true population parameter.

To calculate confidence interval, we use our Standard Error:

If the sampling distribution of the statistic is approximately normal, the 95% confidence interval can be estimated as:
 $(\text{Point Estimate}) \pm (1.96 * \text{Standard Error})$

A confidence interval gives a range of values consistent with the observed data and our estimate of uncertainty.



University of California
San Francisco